

March 2026

DRONE PAYLOAD DESIGN CHALLENGE

Team 2613

OUR CONTENTS

01. Problem Statement

02. Key Challenges

03. Concepts Explored:

04. Final Assembly

05. Working Principle

06. Structural Design and Material

07. Simulation

08. Weight & Payload Validation

09. Electronics

10. Conclusion

PROBLEM STATEMENT

The Challenge

Large-scale reforestation in remote terrain
Manual planting is slow and labour-intensive
Need automated aerial seed dispersal
Must handle non-uniform seedballs (20–35 mm)

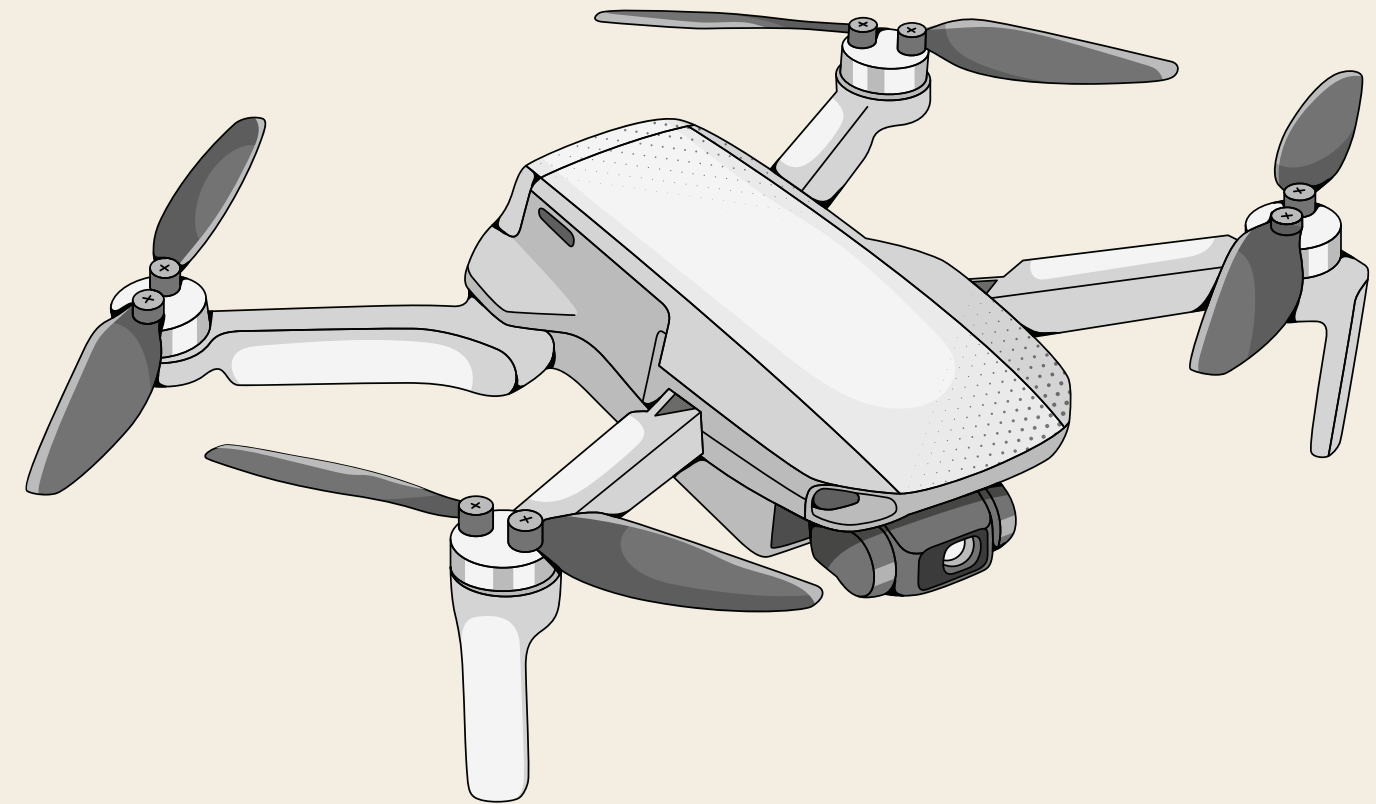
Constraints

Payload limit ≈ 20 kg
Adjustable drop interval 1–5 s
Reliable operation with minimal jamming

DESIGN CHALLENGES

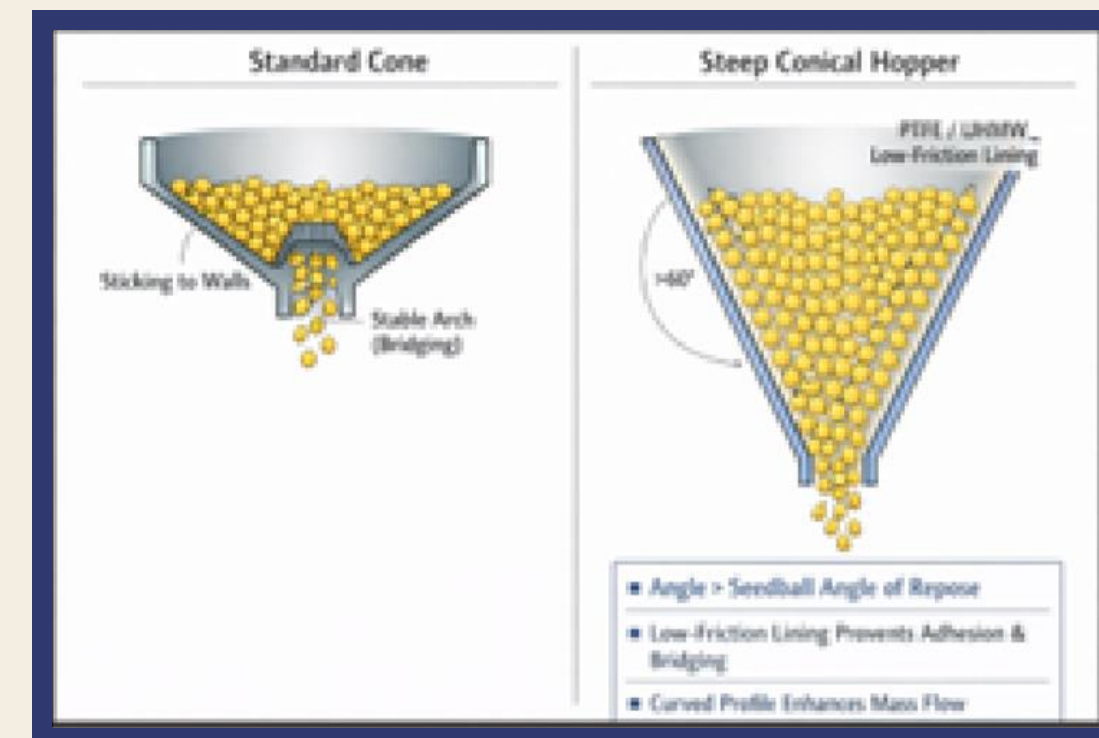
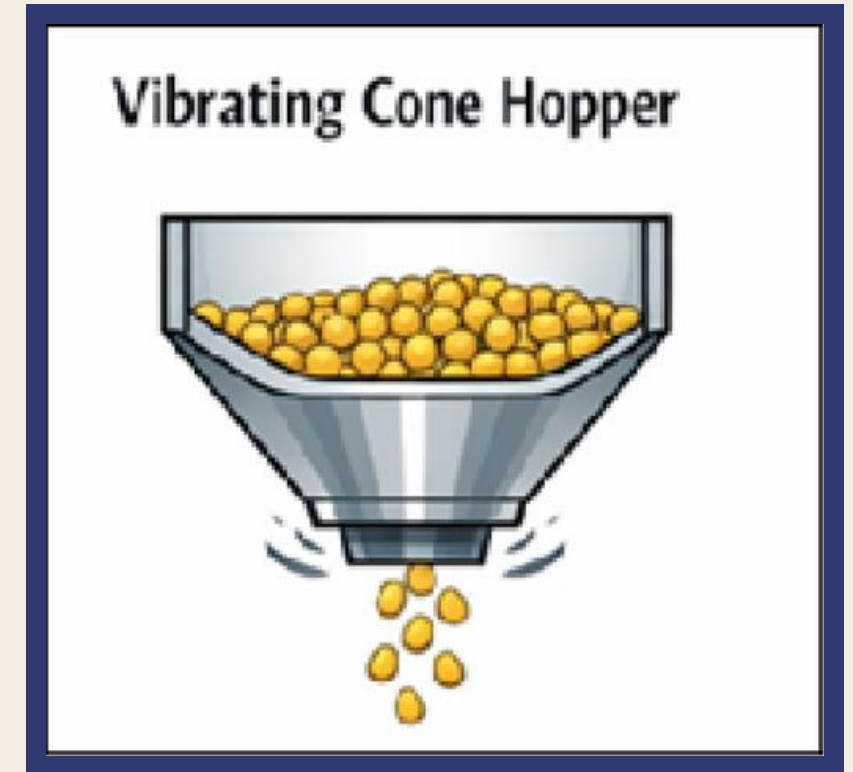
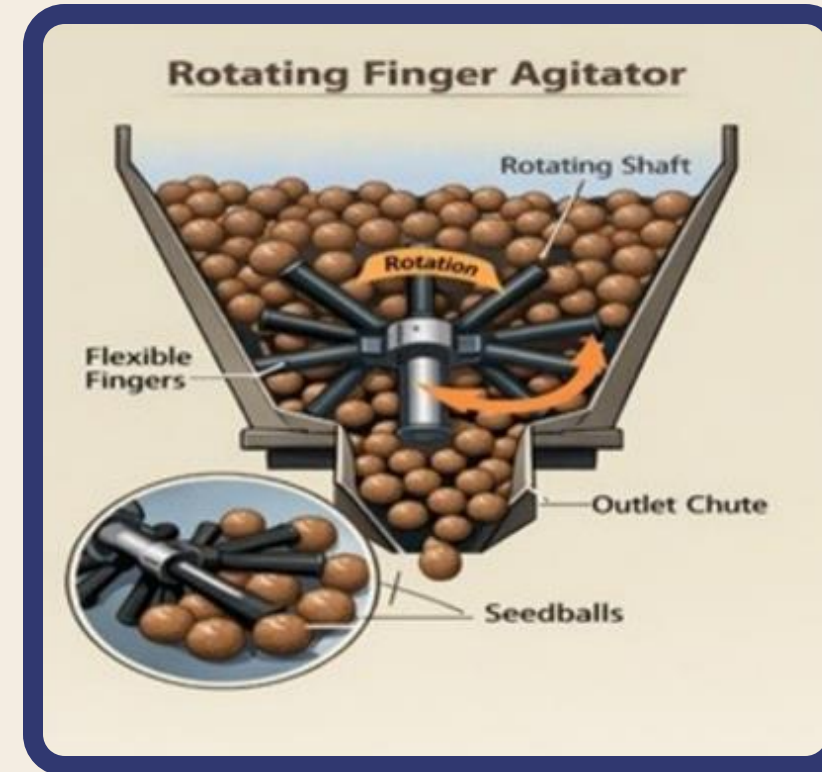
Key Engineering Challenges

- Handling **non-uniform** seedball sizes
- **Preventing jamming** and bridging
- Maintaining drone **stability** during a drop
- Ensuring **controlled timing (1–5 s)**
- Designing a **lightweight** structure under payload limits
- **Operation in dust, vibration and turbulence**



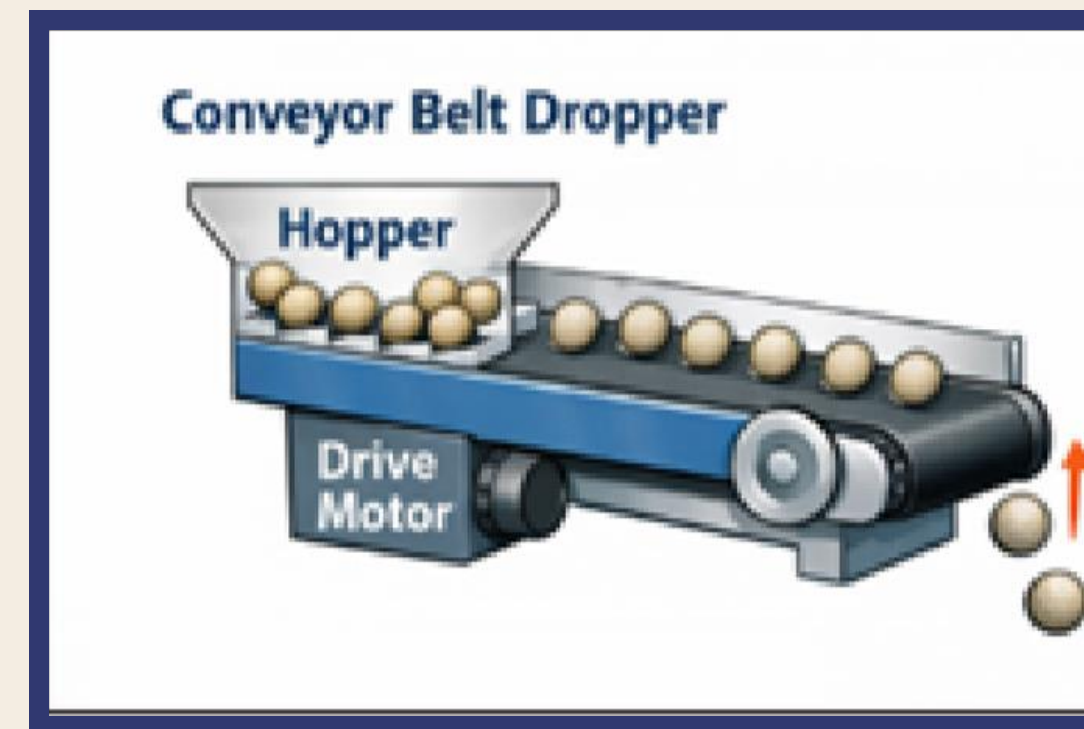
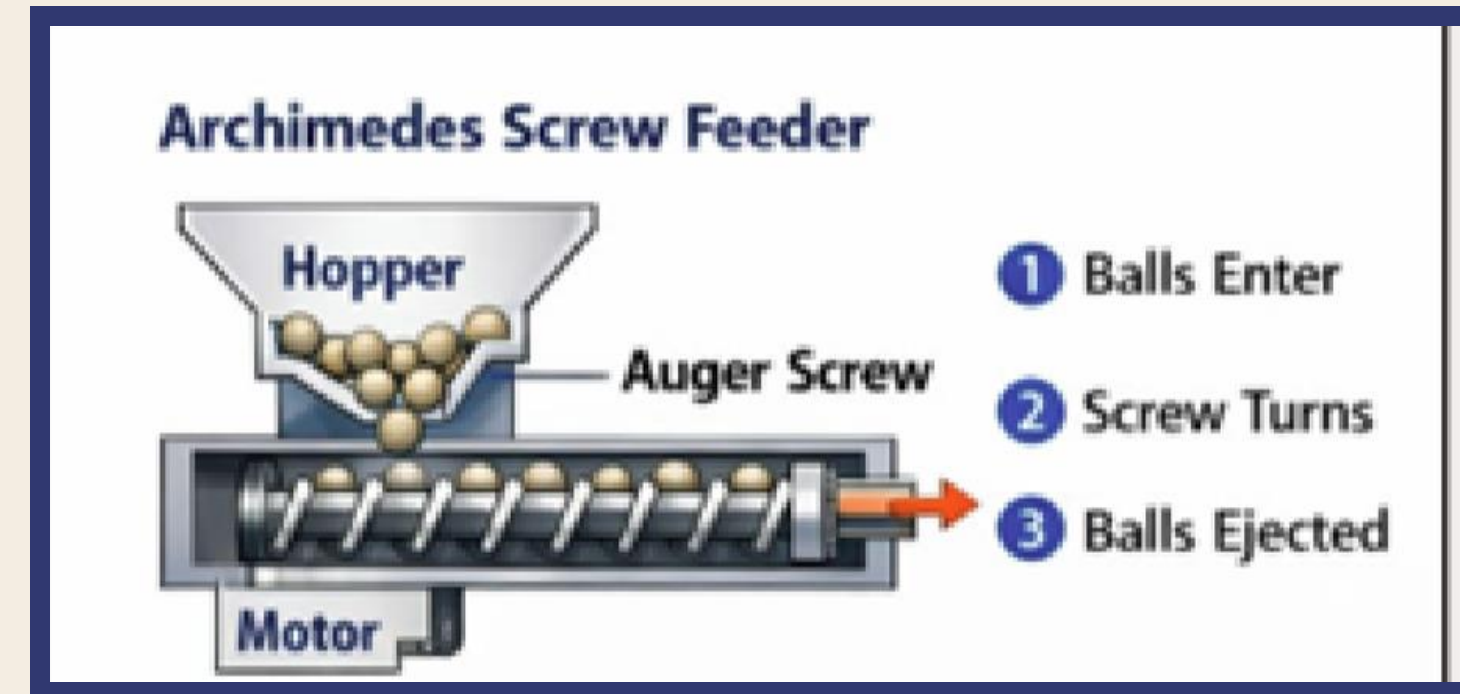
CONCEPTS EXPLORED:

HOPPER CONCEPTS



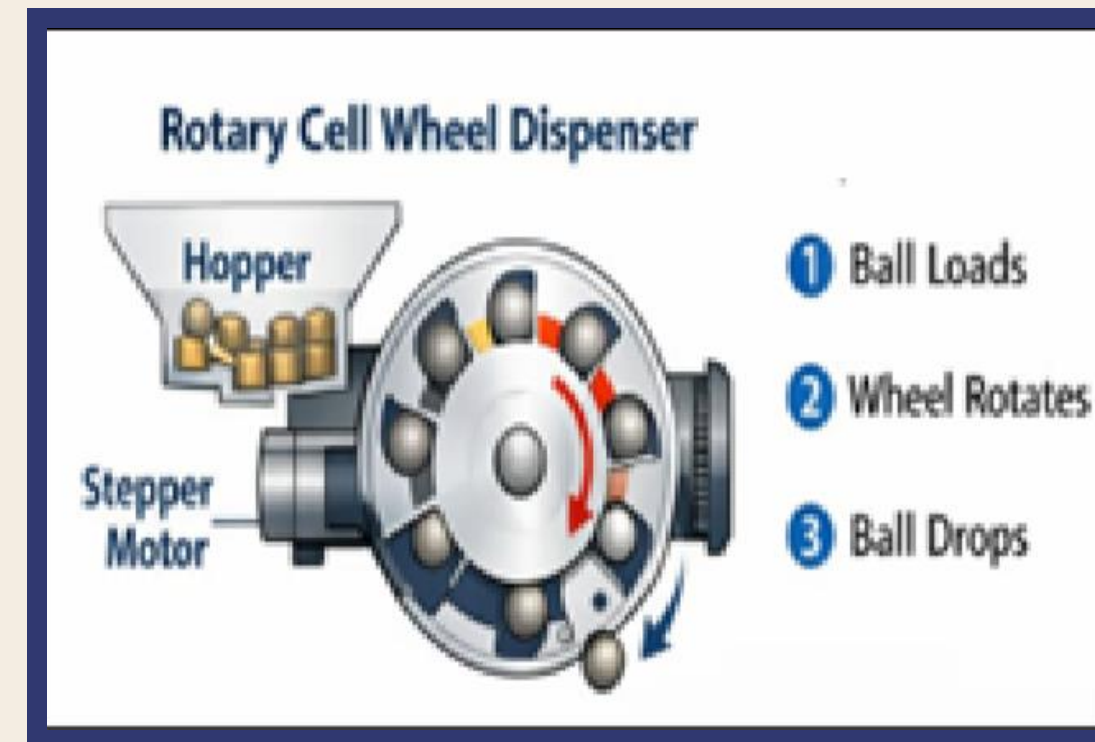
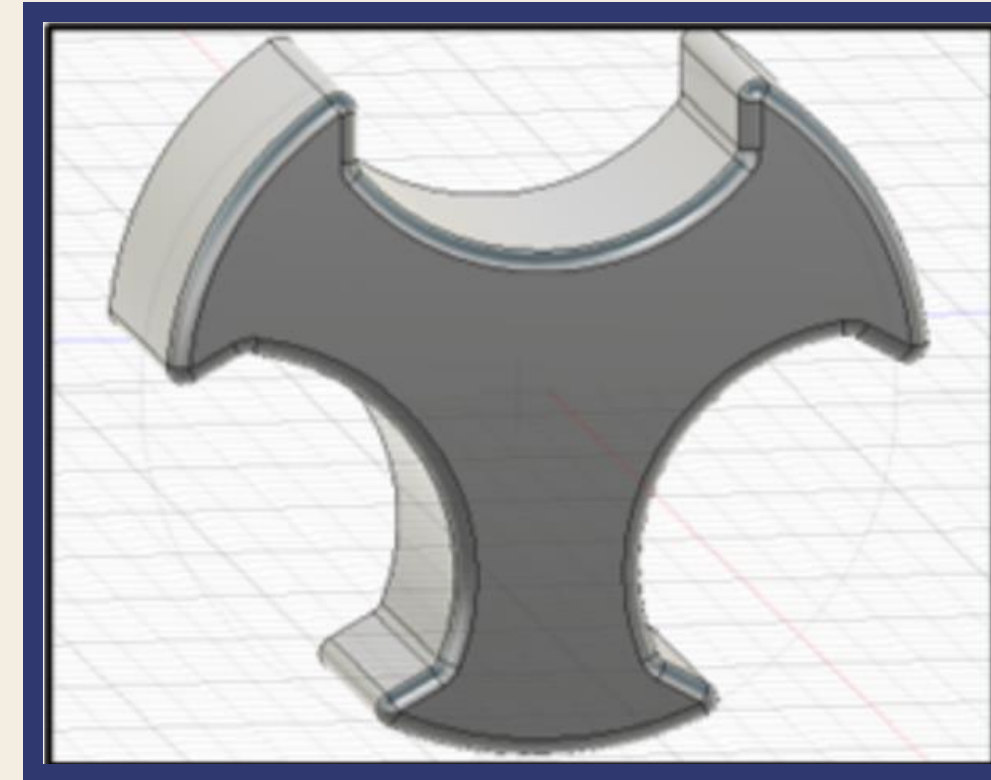
CONCEPTS EXPLORED:

SEPARATION AND FLOW REGULATION

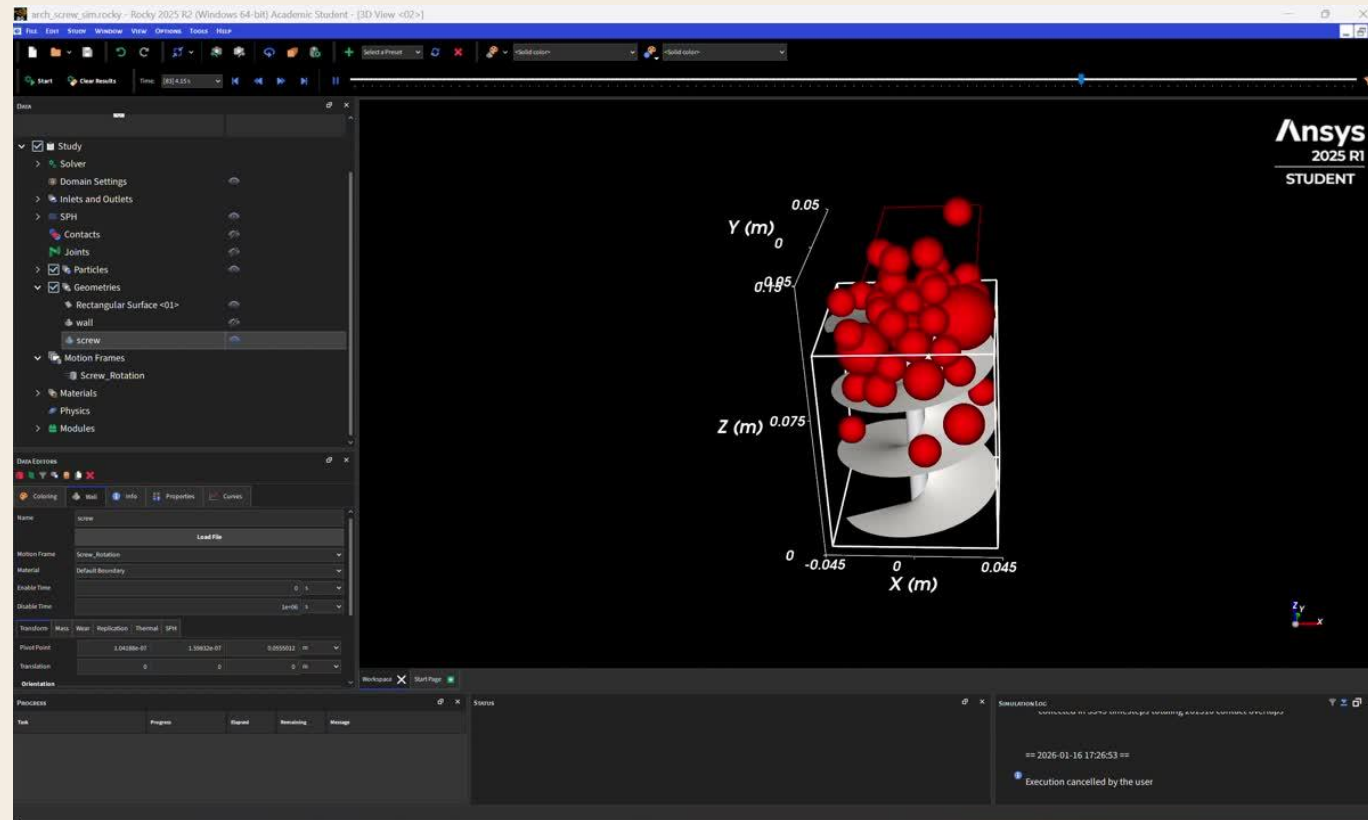


CONCEPTS EXPLORED:

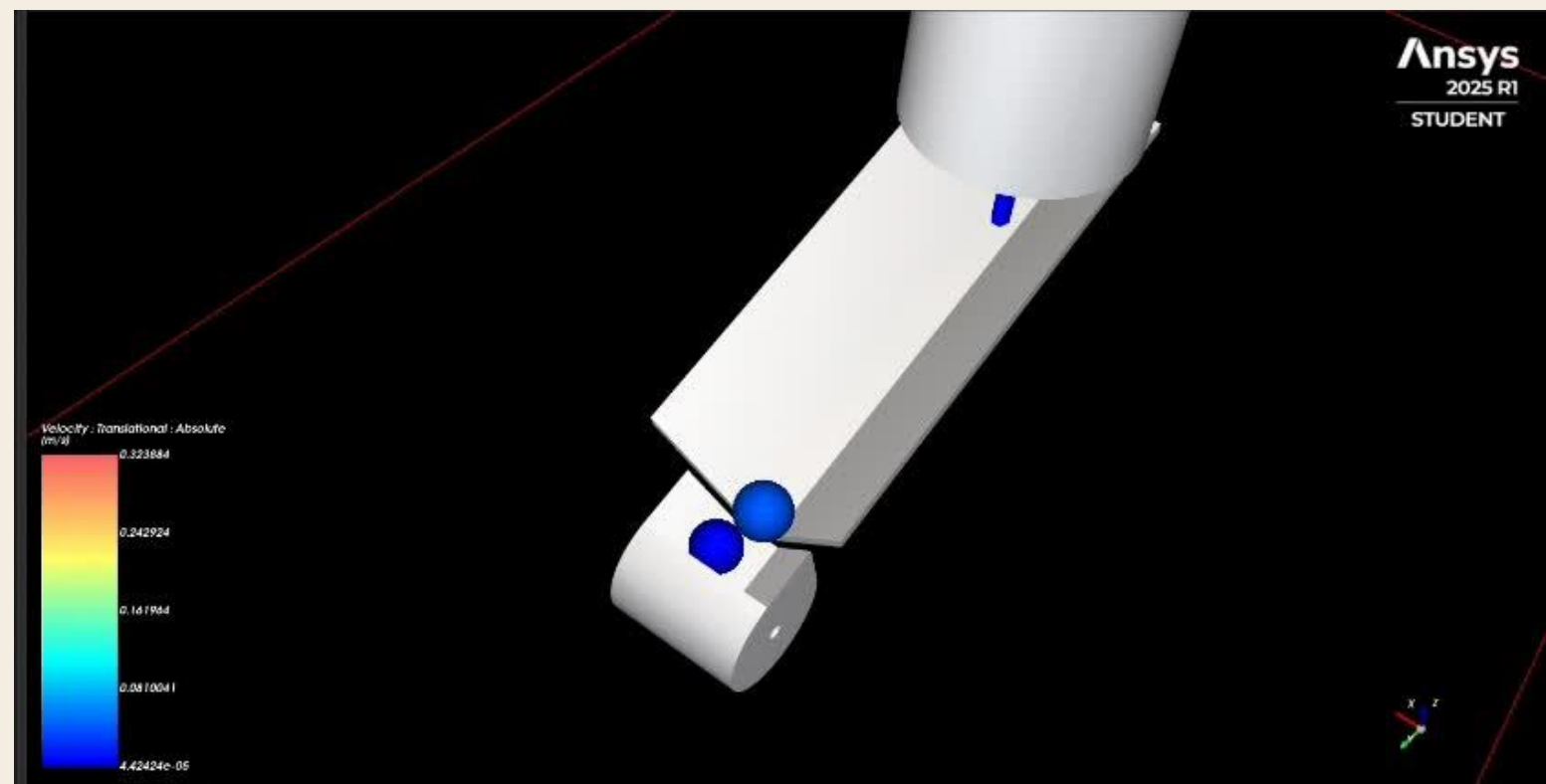
DISPENSING MECHANISM



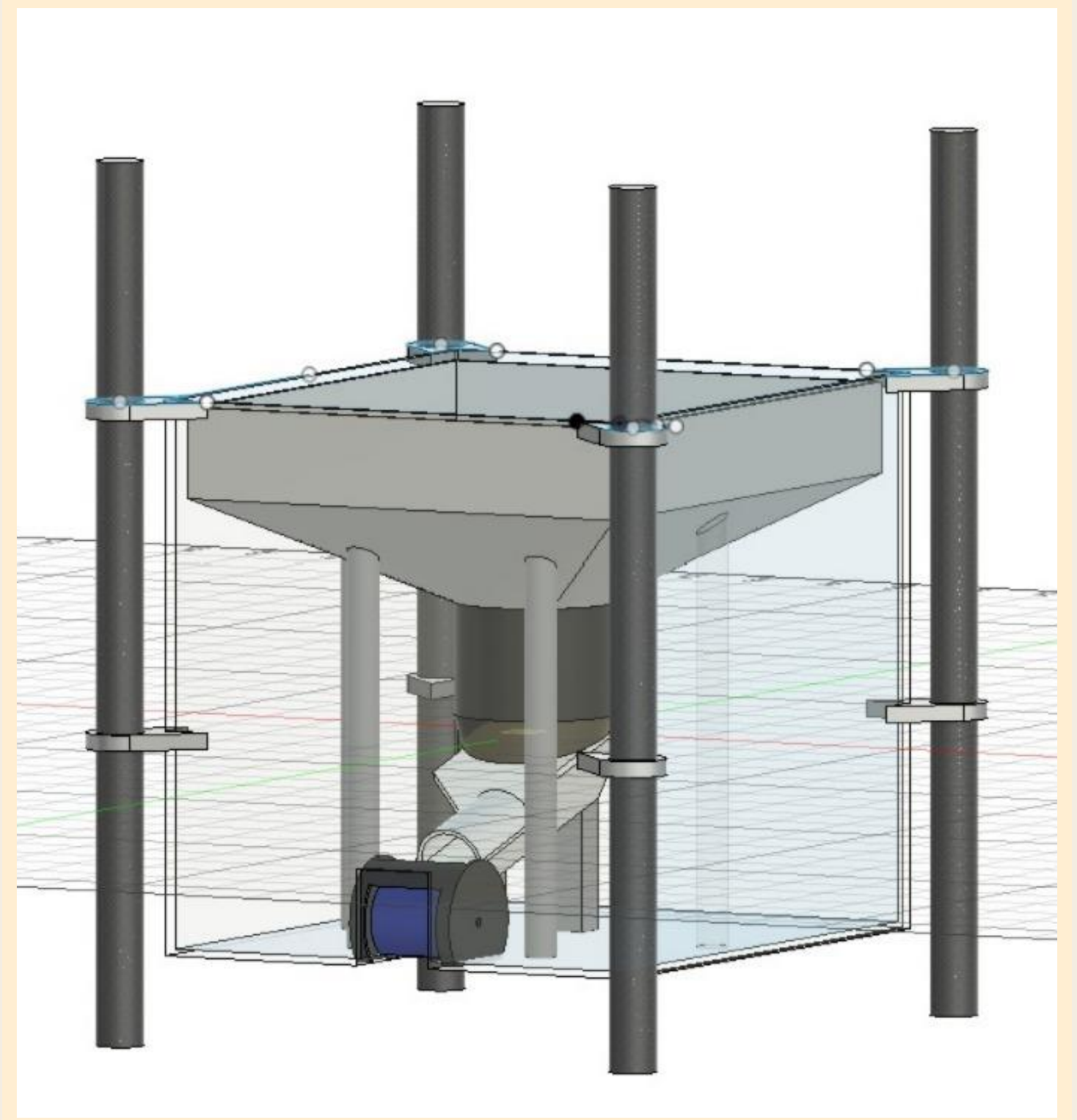
SIMULATION & EVALUATION



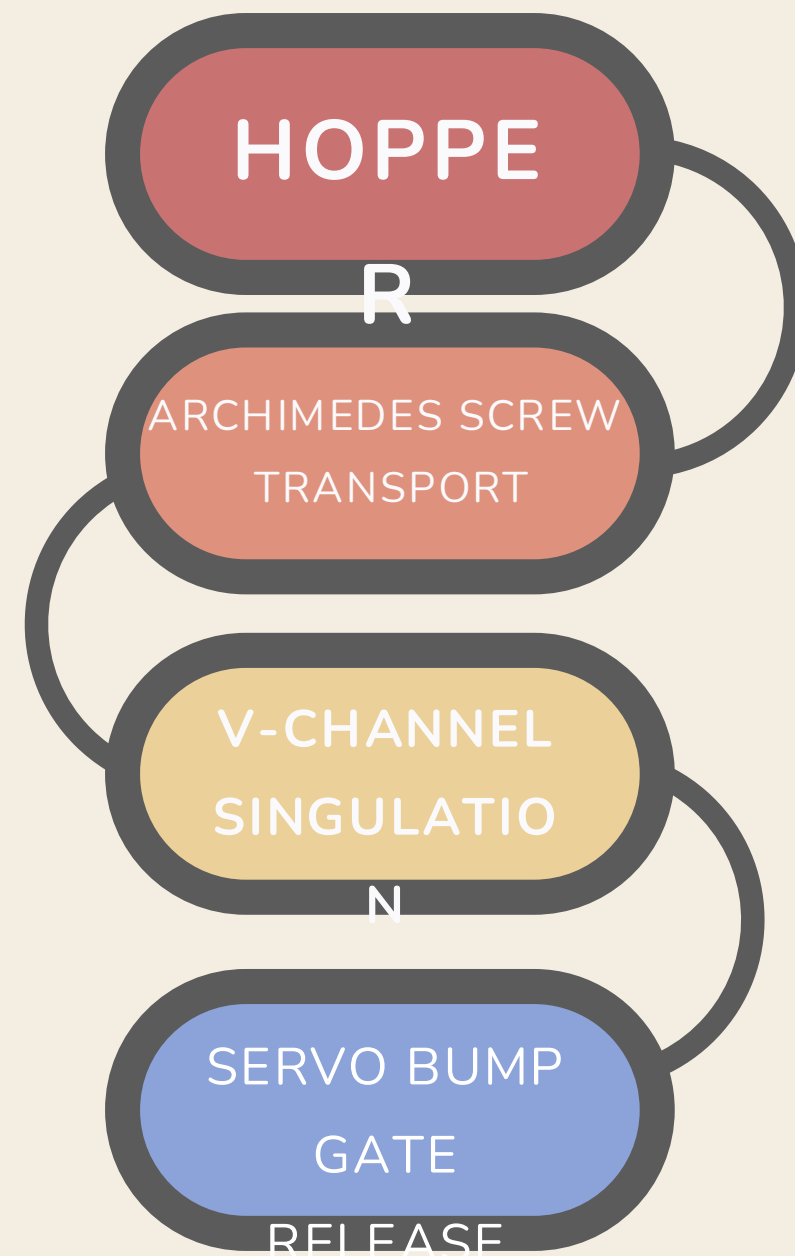
- Discrete Element Method (DEM) used to model seedball–seedball and seedball–wall interactions
- Simulated particle flow through hopper, screw transport, and singulation channel
- Identified bridging, congestion zones, and flow irregularities
- Used results to refine screw clearance, channel geometry, and flow path



FINAL ARCHITECTUR E



WORKING PRINCIPLE



Specs

Single ball release
Adjustable timing
Reduced jamming
probability

STRUCTURAL DESIGN & MATERIALS

Component	Material	S.No.
Funnel	Aluminium	1
Screw & Flow parts	ABS	2
Structures	Aluminium	3

WEIGHT & PAYLOAD VALIDATION

Total payload system mass

Structural components: 3178 g

Actuators: 343 g

Electronics: 190 g

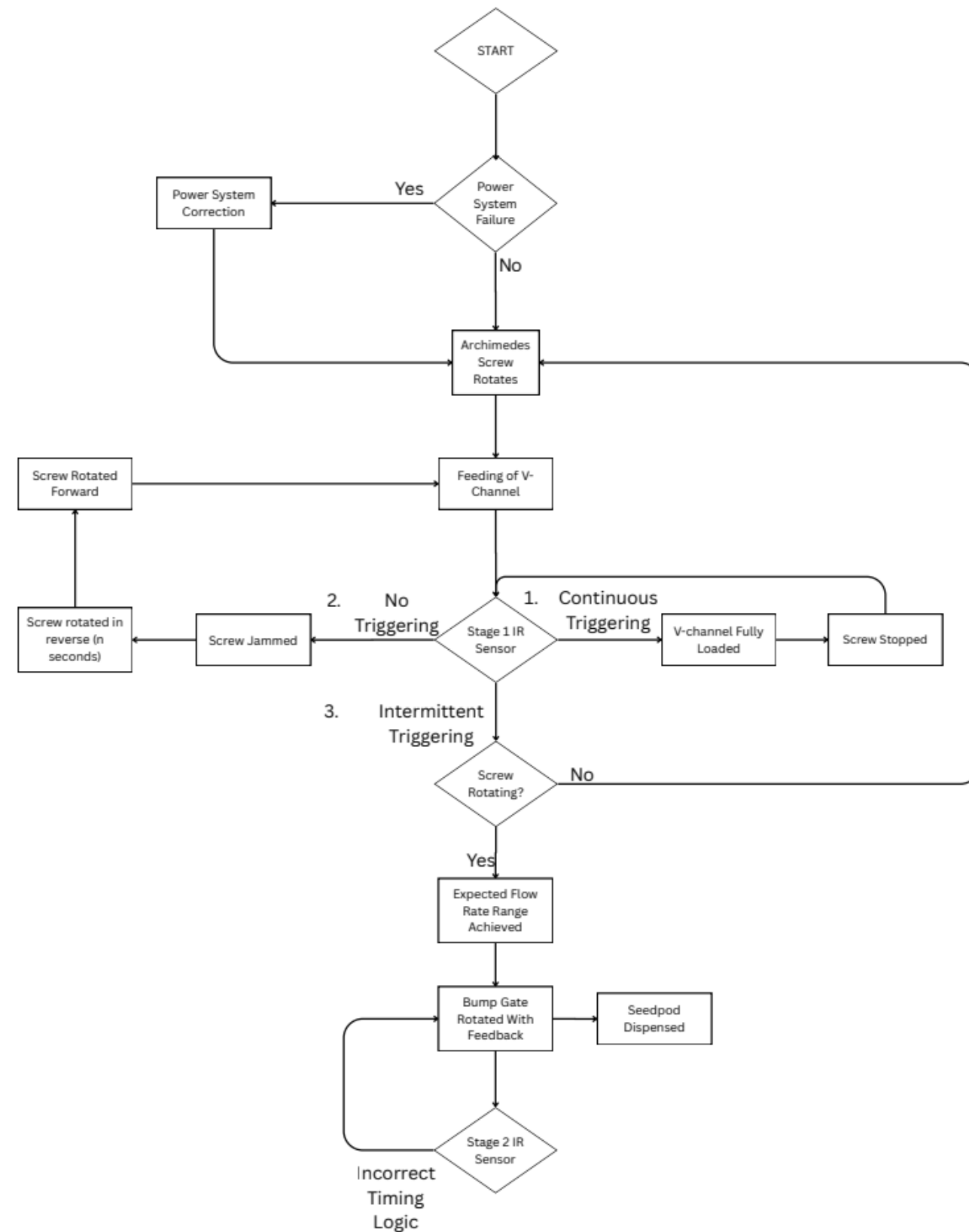
Total dry mass $\approx$ 3.7 kg

Operational payload

~10 kg seedballs

$\approx$ 340 seedballs per load

CONTROL LOGIC



Continuous Mode:

- Archimedes screw stopped
- Bump gate operated

Intermittent Mode:

- Archimedes screw rotates
- Bump gate operated

No Feed Mode:

- Archimedes screw: anti-jamming
- Bump gate closed

FEATURES

- Drop Time Control with IOT
 - The end user could use a HMI or a mobile application to control the rate of seed dispersal wirelessly
- Seedball Counter
 - This gives the user an overall idea of the amount of seed dispersal performed until now
- Non-Blocking Logic
 - This has been utilised to ensure smooth operating of the entire system with seamless sensor integration

CONCLUSION

Key Achievements

Reliable dispensing of 20–35 mm
seedballs

Adjustable drop timing

Autonomous jam mitigation

Lightweight and scalable design

Impact

Enables large-scale aerial
reforestation

Reduces manual labor

Adaptable to different UAV
platforms

